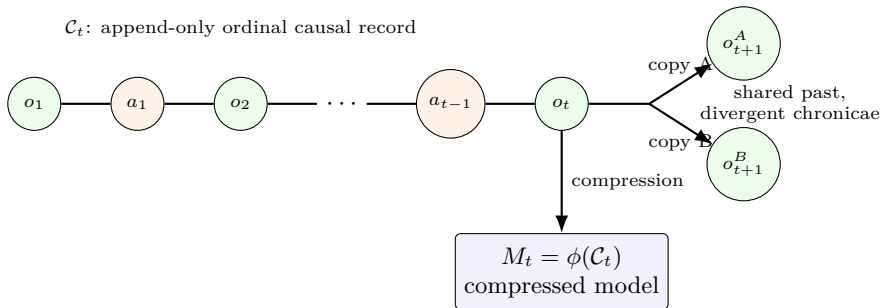


$\mathcal{C}_t$: append-only ordinal causal record



Wall-clock gaps are not represented here unless they enter as events or model content.